

Project Initialization and Planning Phase

Date	27 June 2026
Team ID	XXXXXXX
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to improve the credit card approval process using machine learning by providing accurate and fast approval predictions. The system reduces manual effort, minimizes human errors, and helps financial institutions make reliable decisions. The application provides a user-friendly interface for entering applicant details and obtaining instant prediction results.

Project Overview	
Objective	The proposal report aims to automate the credit card approval process using machine learning. It improves prediction accuracy, reduces processing time, minimizes manual effort, and provides reliable approval decisions through a web-based application.
Scope	The project collects applicant information, preprocesses the input data, predicts credit card approval using a trained machine learning model, and displays the prediction result through a Flask web application.
Problem Statement	
Description	Financial institutions often rely on manual verification and rule-based systems for credit card approval. This process is time-consuming, inconsistent, and may lead to incorrect decisions. An automated prediction system can improve efficiency and accuracy.
Impact	The proposed solution reduces approval time, improves decision accuracy, minimizes manual work, and provides faster and more consistent credit card approval predictions.
Proposed Solution	

Approach	Use Python, Flask, Machine Learning, and data preprocessing techniques to analyze applicant information and predict credit card approval. The trained model is integrated into a web application that provides instant prediction results.
Key Features	- Machine learning-based credit card approval prediction. - User-friendly Flask web application. - Automated data preprocessing. - Real-time prediction results. - Fast and reliable decision support. - Simple and responsive interface.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Computing Resources	Intel Core i5 Processor
Memory	RAM specifications	8 GB
Storage	Disk space for application, models, and logs	256 GB SSD
Software		
Frameworks	software	Python, Flask
Libraries	Additional Libraries	Scikit-learn, Pandas, NumPy, Joblib, Flask, Unicorn
Development Environment	IDE	Visual Studio Code
Data		
Data	Source, size, format	Credit card applicant dataset, trained Random Forest model (.joblib), JSON prediction history (prediction_summary.json)